

Lesson 2: Intro to data & numerical summaries

TB sections 1.2, 1.4

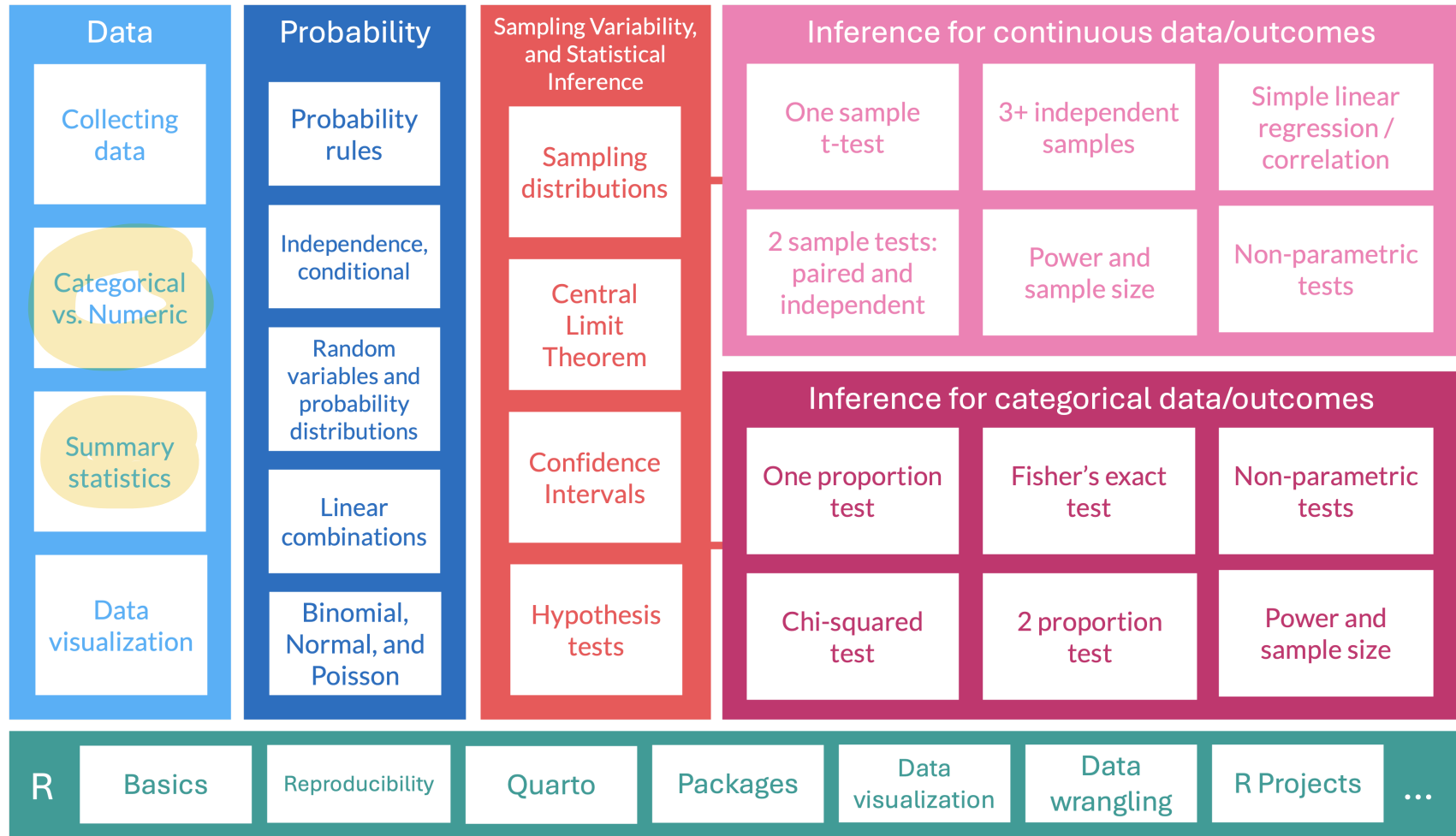
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2025-10-01

Learning Objectives

1. Define observations and variables, and recognize them in a data frame.
2. Define four variable types in data.
3. Define and calculate measures of center (including mean and median).
4. Define and calculate measures of spread (including standard deviation and interquartile range).

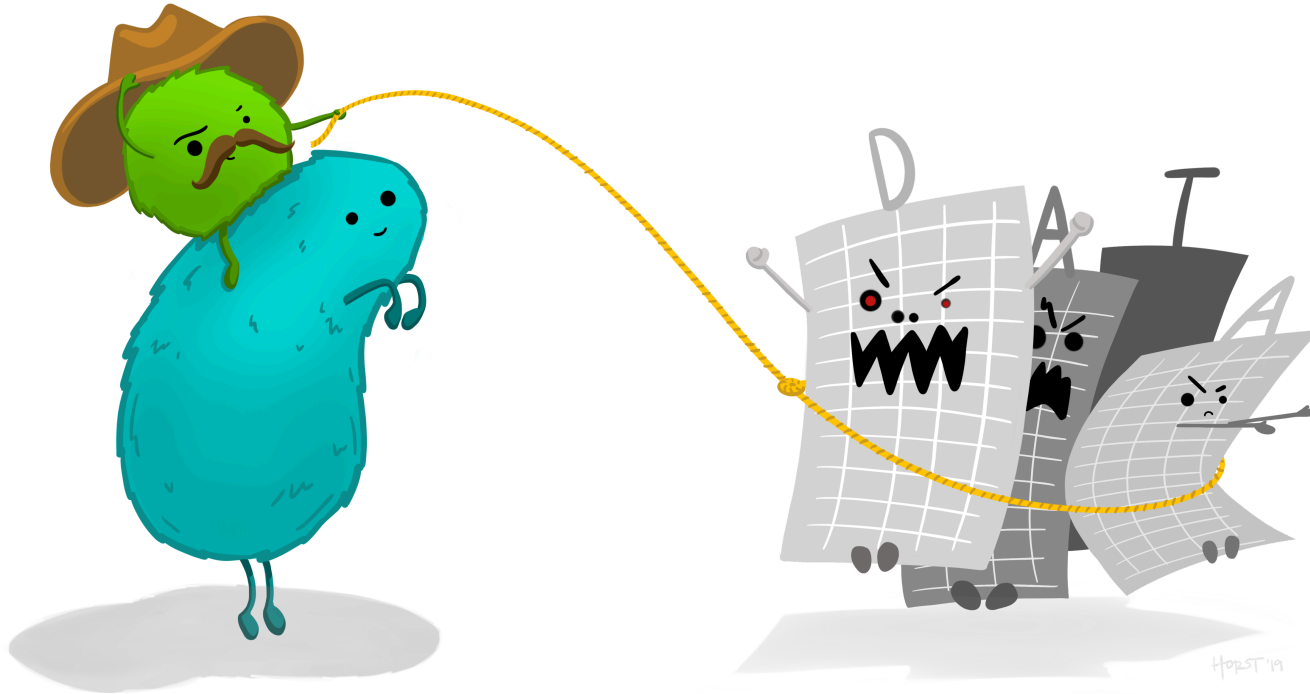
Where are we?



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Intro to Data



Artwork by @allison_horst

Example: the frog study¹

In evolutionary biology, parental investment refers to the amount of time, energy, or other resources devoted towards raising offspring.

We will be working with the frog dataset, which originates from a 2013 study² about maternal investment in a frog species. Reproduction is a costly process for female frogs, necessitating a trade-off between individual egg size and total number of eggs produced.

Researchers were interested in investigating how maternal investment varies with altitude. They collected measurements on egg clutches found at breeding ponds across 11 study sites; for 5 sites, the body size of individual female frogs was also recorded.

Poll Everywhere Question 1

14:00 Wed Oct 1

Join by Web PollEv.com/nickywakim275



Review from last class: What is the target population for the frog study?

female frogs

Parental frogs

All maternal frogs that bear egg clutches

female-frogs

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female frogs
w/ eggs

Four rows from frog data frame

	<u>altitude</u>	latitude	egg.size	clutch.size	clutch.volume	^{frog} body.size
1	3,462.00	34.82	1.95	181.97	177.83	3.63
2	3,462.00	34.82	1.95	269.15	257.04	3.63
3	3,462.00	34.82	1.95	158.49	151.36	3.72
150	<u>2,597.00</u>	<u>34.05</u>	<u>2.24</u>	<u>537.03</u>	<u>776.25</u>	<u>NA</u>

Four rows from frog data frame

	altitude	latitude	egg.size	clutch.size_ <i>mm</i>	clutch.volume	body.size
1	3,462.00	34.82	1.95	181.97	177.83	3.63
2	3,462.00	34.82	1.95	269.15	257.04	3.63
3	3,462.00	34.82	1.95	158.49	151.36	3.72
150	2,597.00	34.05	2.24	537.03	776.25	NA

- Each **row** is an **observation**
- Each **column** is a **variable**
- All the **observations** and **variables** together make a **data frame** (sometimes called data matrix)
- **Missing values:** **NA** means the measured value for body size in clutch #150 is missing

Frog study: variables and their descriptions

- **Variables** are recorded characteristics for each **observation**

Variable	Description
altitude	Altitude of the study site in meters above sea level
latitude	Latitude of the study site measured in degrees
egg.size	Average diameter of an individual egg to the 0.01 mm
clutch.size	Estimated <u>number of eggs</u> in clutch
clutch.volume	Volume of egg clutch in mm ³
body.size	Length of egg-laying frog in cm

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Types of variables (1/2)

Numerical variables

Numerical variables take on numerical values, such that numerical operations (sums, differences, etc.) are reasonable.

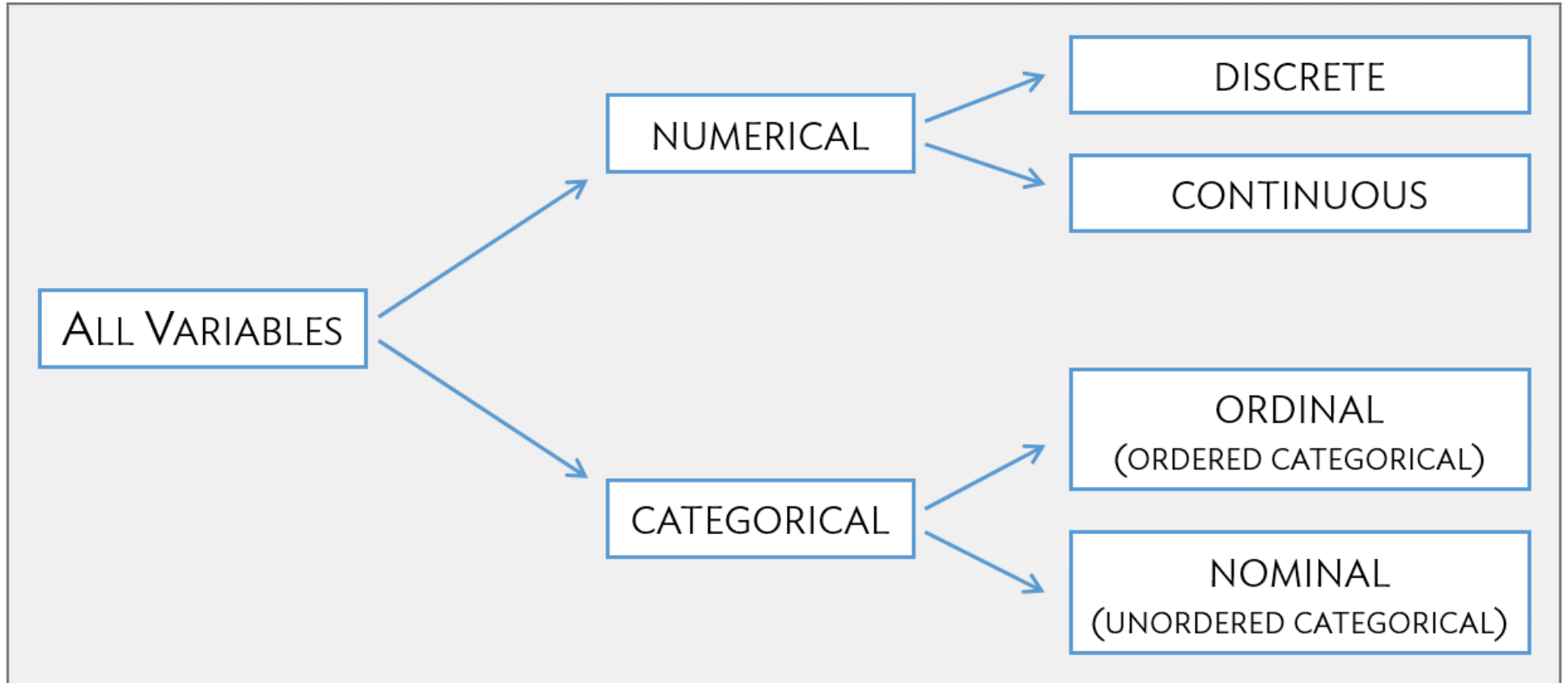
- **Discrete:** only take on integer values (e.g., # of family members)
- **Continuous:** can take on any value within a specified range (e.g., height)

Categorical variables

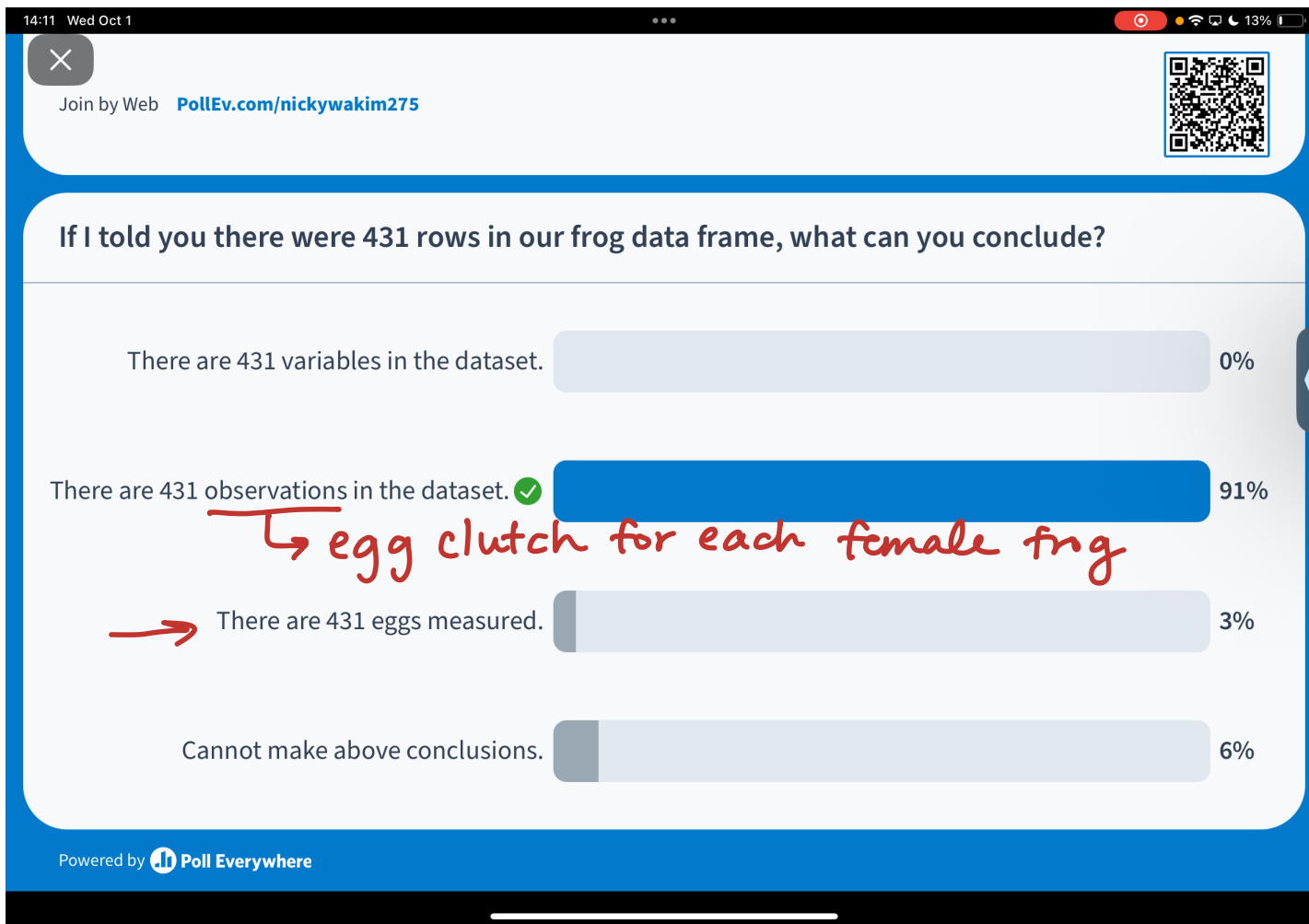
Categorical variables take on values that are names or labels; the possible values are called the variable's *levels*.

- **Ordinal:** exists some natural ordering of levels (e.g., level of education)
- **Nominal:** no natural ordering of levels (e.g., gender identity)

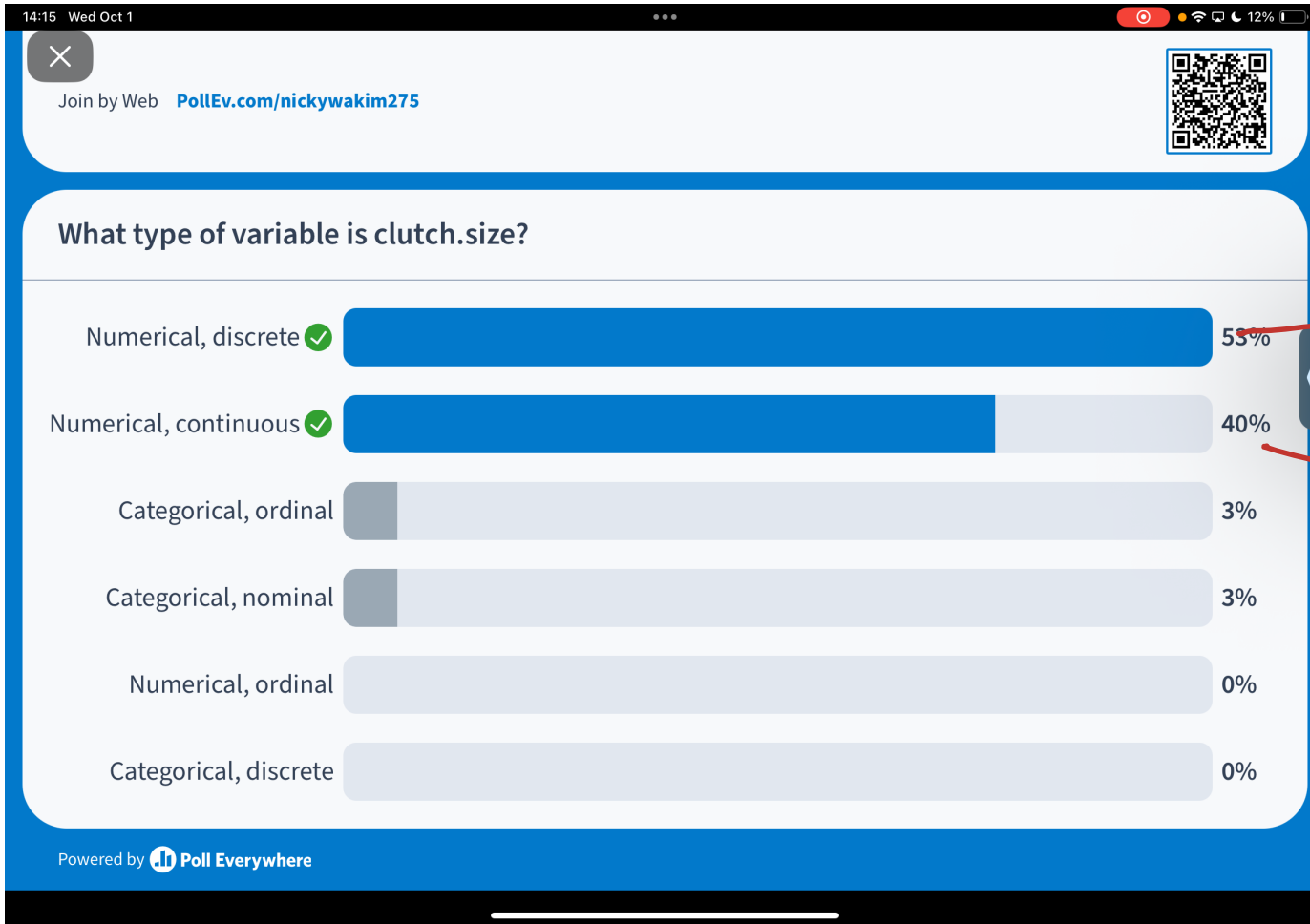
Types of variables (2/2)



Poll Everywhere Question 2



Poll Everywhere Question 3



→ based on description

→ based on the actual data

Variable (column) types in R

- We have not done much with R yet, but I want this to serve as a reference for you!
- Variable types (as we speak them) are translated by R a little differently
- Below is the mapping of R types to variable types

R type	Variable type	Description
integer	discrete	integer-valued numbers
double or numeric	continuous	numbers that are decimals
<u>factor</u> ★ best	<u>categorical</u>	categorical variables stored with <u>levels (groups)</u>
<u>character</u>	categorical	<u>text, "strings"</u>
logical	categorical	boolean (TRUE, FALSE)

Exploring data initially

- Techniques for exploring and summarizing data **differ** for **numerical** versus **categorical** variables.
- Numerical and graphical summaries are useful for examining variables one at a time
 - Can also be used for exploring the relationships between variables
 - *Numerical* summaries are not just for **numerical** variables (certain ones are used for **categorical** variables)
- Let's start looking into ways to summarize and explore **numerical** data!
 - We will come back to **categorical** variables another day

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Warning!

I decided to keep some R code in these slides. It's going to be a little confusing now, but I thought it would be a worthwhile reference as soon as we get through R basics

Measures of center: mean

$1.5e-7$
 1.5×10^{-7}

Sample mean

the average value of observations

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \sum_{i=1}^n \frac{x_i}{n}$$

where $x_1, x_2, \dots, x_n$ represent the n observed values in a sample

Example: What is the mean clutch volume in the **frog** dataset?

$$\bar{x} = \frac{\sum_{i=1}^{431} x_i}{431}$$

$$\frac{\sum_{i=1}^{431} \text{clutch vol}_i}{431}$$

```
1 mean(frog$clutch.volume, na.rm = T)  
[1] 882.474
```

Answer: the mean clutch volume is 882.5 mm³.

Measures of center: median

Median

- The middle value of the observations in a sample
- The median is the 50th percentile, meaning
 - 50% of observations lie below the median
 - 50% of observations lie above the median
- If the number of observations is
 - Odd: the median is the middle observed value
 - Even: the median is the average of the two middle observed values
- We can calculate the median clutch volume

```
1 median(frog$clutch.volume)
```

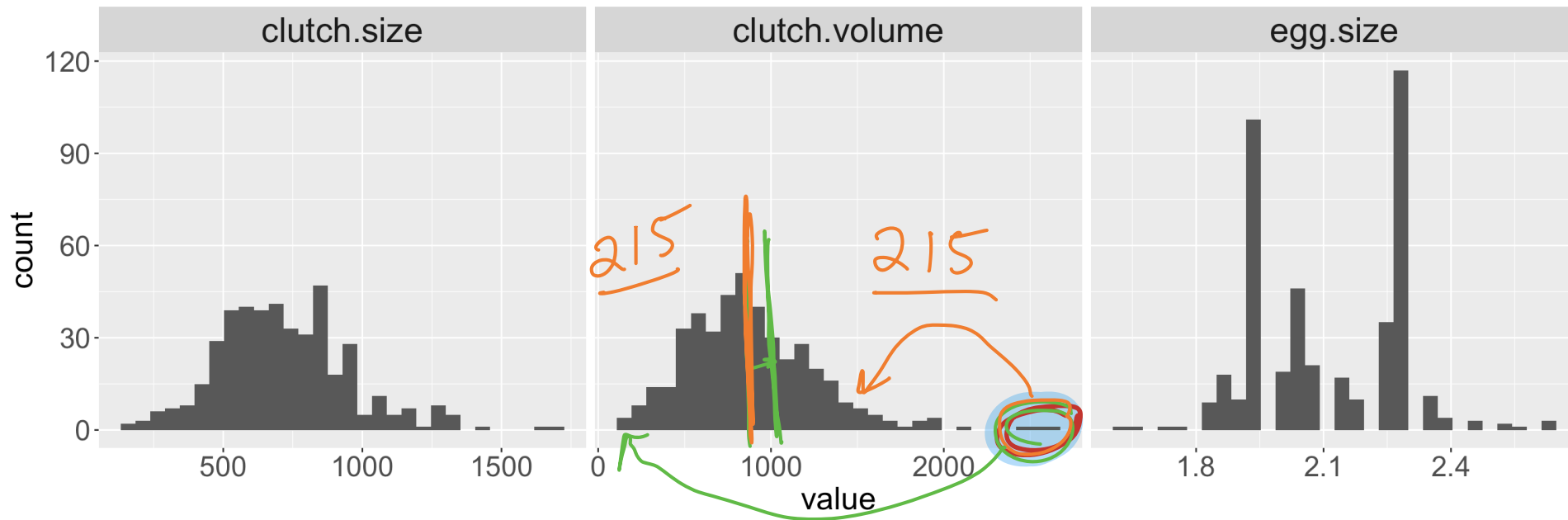
```
[1] 831.7638
```

2 | 6th
lowest

831.8 mm³

Measures of center: mean vs. median

Histograms of three variables in frog dataset



- Mean values will be pulled towards extreme values

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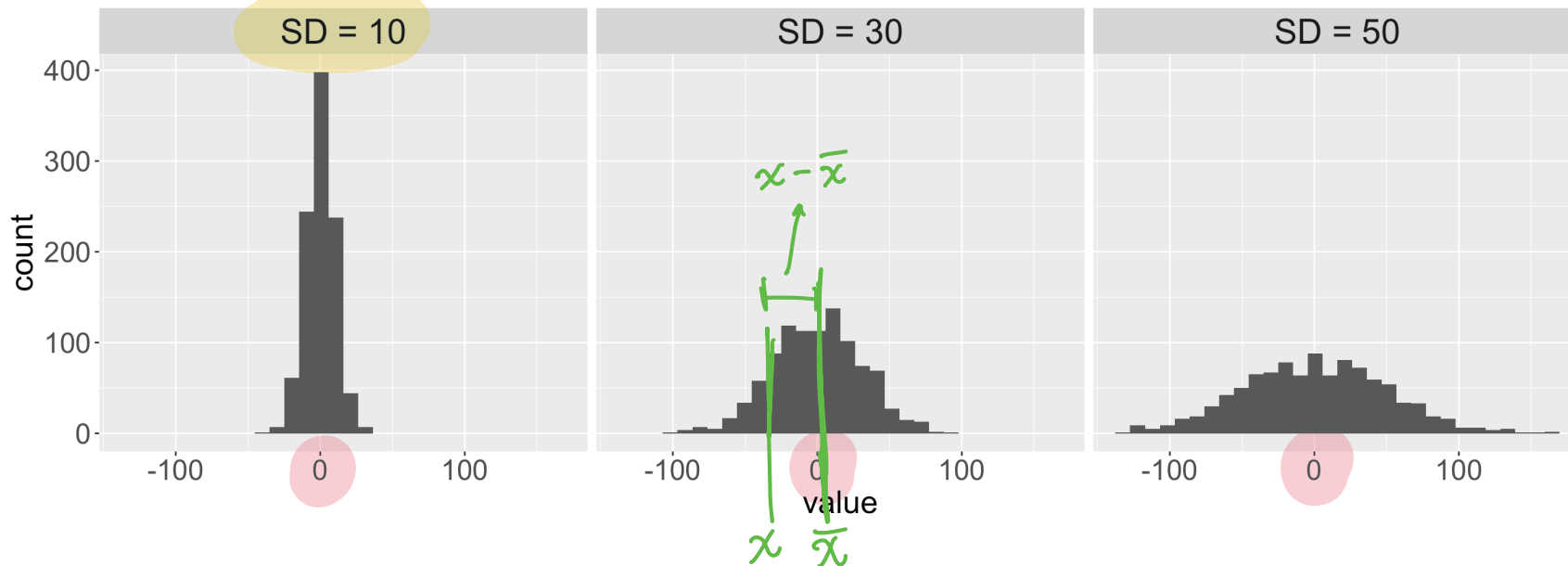
Measures of spread: standard deviation (SD) (1/3)

Standard deviation (SD)

(Approximately) the average distance between an observation and the mean

- An observation's **deviation** is the distance between its value x and the sample mean $\bar{x}$: deviation = $x - \bar{x}$

Simulated data with different standard deviations



Measures of spread: SD (2/3)

- The **sample variance** s^2 is the sum of squared deviations divided by the number of observations minus 1.

$$s^2 = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1} = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n - 1}$$

where $x_1, x_2, \dots, x_n$ represent the n observed values.

- The **standard deviation** s (or *sd*) is the square root of the variance.

$$s = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1}} = \sqrt{\sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n - 1}}$$

↓
variance

Measures of spread: SD (3/3)

Let's calculate the sample standard deviation for the clutch volume:

$$s = \sqrt{\sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}} =$$

- Doing this by hand would be really time consuming!
- R can easily do this for us!

```
1 mean(frog$clutch.volume)
```

```
[1] 882.474
```

```
1 sd(frog$clutch.volume)
```

```
[1] 379.0527
```

Answer: The standard deviation of the clutch volume is ~~379.05~~ mm³

Variance
379.05² (mm³)²

Empirical Rule: one way to think about the SD

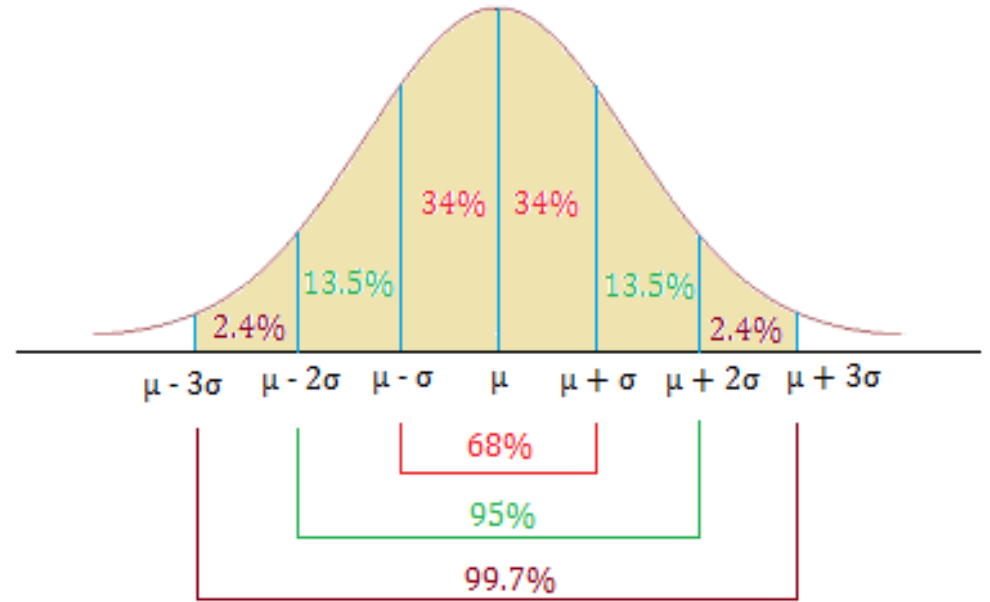
For symmetric bell-shaped data, about

- 68% of the data are within 1 SD of the mean
- 95% of the data are within 2 SD's of the mean
- 99.7% of the data are within 3 SD's of the mean

These percentages are based off of percentages of a true normal distribution.

$$882 \pm 380$$

$$882 \pm 2 \times 380$$



Empirical rule

<https://statistics-made-easy.com/empirical-rule/>

Measures of spread: interquartile range (IQR) (1/2)

The p^{th} percentile is the observation such that $p\%$ of the remaining observations fall below this observation.

- The *first quartile* Q_1 is the 25th percentile. *lowest value to 25th % lowest value*
- The *second quartile* Q_2 , i.e., the median, is the 50th percentile.
- The *third quartile* Q_3 is the 75th percentile.

Interquartile range (IQR)

The distance between the third and first quartiles.

$$IQR = Q_3 - Q_1$$

- IQR is the width of the *middle half* of the data

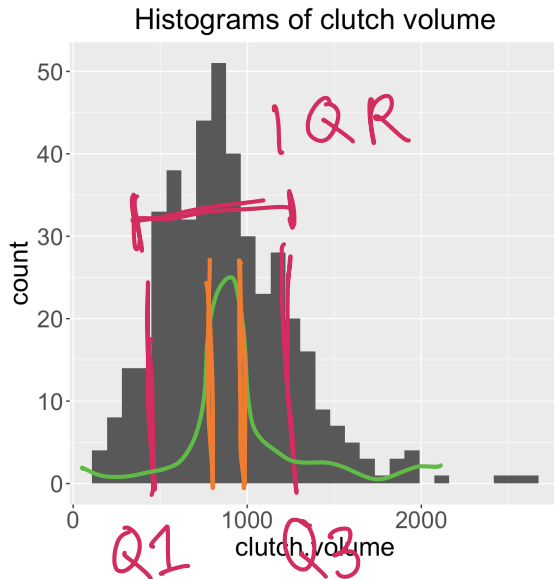
Measures of spread: IQR (2/2)

5 number summary

```
1 summary(frog$clutch.volume)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
151.4	609.6	831.8	882.5	1096.5	2630.3

$$IQR = Q_3 - Q_1 = 1096.5 - 609.6 = 486.9$$



What is the IQR of the sepal widths?

```
1 quantile(frog$clutch.volume, c(.25, .75))
```

25%	75%
609.5773	1096.4782

```
1 diff(quantile(frog$clutch.volume, c(.25, .75)))
```

75%
486.9009

```
1 IQR(frog$clutch.volume)
```

```
[1] 486.9009
```

Robust estimates

Summary statistics are called **robust estimates** if extreme observations have little effect on their values

Estimate	Robust?
Sample mean	✗
Median	✓
Standard deviation	✗
IQR	✓

- For samples with many extreme values, the median and IQR might provide a more accurate sense of the center and spread

Poll Everywhere Question 4

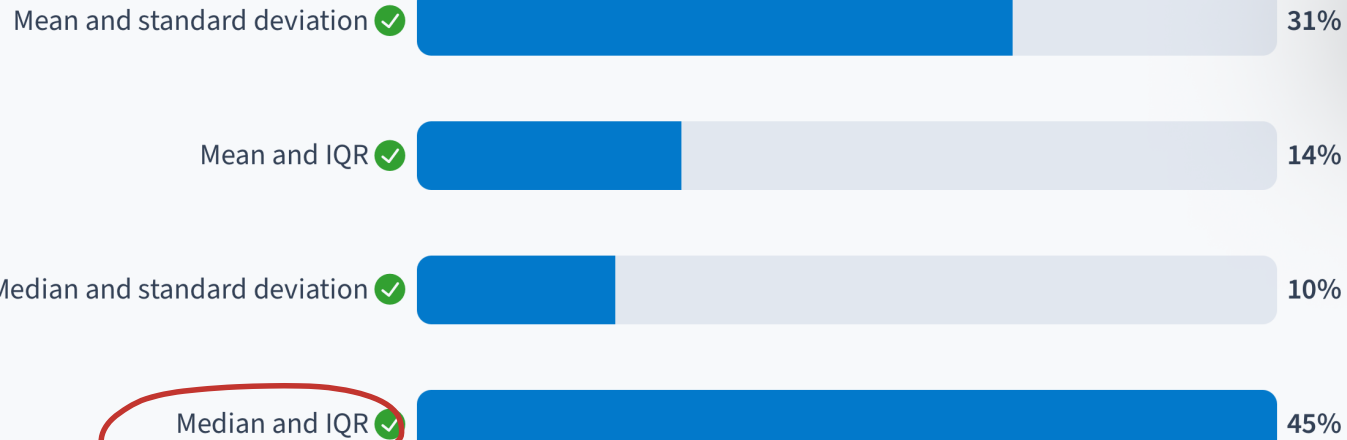
14:46 Wed Oct 1



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Which combination of measures of center and spread would you use for the clutch volume?



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→ bc slightly skewed